

PAC2507-0010

Leaflet_KLF0030S1_Covastin Tablet

Size: 148(L) x 210(H)mm

Colors: Black

Current Revision: 002

Updated Revision: 003

Important information. Please read carefully.

COVASTIN®
(10mg, 20mg & 40mg)

COMPOSITION

Covastin Tablet 10mg: Each film coated tablet contains Simvastatin 10mg
Covastin Tablet 20mg: Each film coated tablet contains Simvastatin 20mg
Covastin Tablet 40mg: Each film coated tablet contains Simvastatin 40mg

PHARMACODYNAMICS

Mechanism of action

After oral ingestion, simvastatin which is an inactive lactone, is hydrolysed in the liver to the corresponding active beta-hydroxyacid form which has a potent activity in inhibiting HMG-CoA reductase (3 hydroxy – 3 methylglutaryl CoA reductase). This enzyme catalyses the conversion of HMG-CoA to mevalonate, an early and rate-limiting step in the biosynthesis of cholesterol.

Simvastatin has been shown to reduce both normal and elevated LDL-C concentrations. LDL is formed from very-low-density protein (VLDL) and is catabolised predominantly by the high affinity LDL receptor. The mechanism of the LDL-lowering effect of simvastatin may involve both reduction of VLDL-cholesterol (VLDL-C) concentration and induction of the LDL receptor, leading to reduced production and increased catabolism of LDL-C. Apolipoprotein B also falls substantially during treatment with simvastatin. In addition, simvastatin moderately increases HDL-C and reduces plasma TG. As a result of these changes the ratios of total- to HDL-C and LDL- to HDL-C are reduced.

PHARMACOKINETICS

Simvastatin is absorbed from the gastrointestinal tract and must be hydrolysed to its active β -hydroxyacid form. Other active metabolites have been detected and several inactive metabolites are also formed. Simvastatin is a substrate for the cytochrome P450 isoenzyme CYP3A4 and undergoes extensive first-pass metabolism in the liver; its primary site of action. Less than 5% of the oral dose has been reported to reach the circulation as active metabolites. Both simvastatin and its β -hydroxyacid metabolite are about 95 % bound to plasma proteins. Simvastatin is mainly excreted in the faeces via the bile as metabolites. About 70 to 15 % is recovered in the urine, mainly in inactive forms. The half-life of the active β -hydroxyacid metabolite is 1.9 hours.

INDICATIONS

Reductions in Risk Of CHD Mortality And Cardiovascular Events

In patients at high risk of coronary events because of existing coronary heart disease, diabetes, peripheral vessel disease, history of stroke or other cerebrovascular disease, simvastatin is indicated to:

- Reduce the risk of total mortality by reducing CHD deaths.
- Reduce the risk of non-fatal myocardial infarction and stroke.
- Reduce the need for coronary and non-coronary revascularisation procedures.

Hyperlipidaemia

Simvastatin is indicated as an adjunct to diet to reduce elevated total-C, LDL-C, Apo B, and TG, and to increase HDL-C in patients with primary hypercholesterolemia, heterozygous familial hypercholesterolemia or combined (mixed) hyperlipidaemia when response to diet and other non-pharmacological measures is inadequate. Simvastatin therefore, lowers the LDL-C/HDL-C and the total-C/HDL-C ratios.

Pediatric Patients With Heterozygous Familial Hypercholesterolemia

Simvastatin is indicated as an adjunct to diet to reduce total-C, LDL-C, TG, and Apo B levels in adolescent boys and girls who are at least one year post-menarche, 10-17 years of age, with heterozygous familial Hypercholesterolemia (HeFH).

General Recommendations

Prior to initiating therapy with Simvastatin, secondary causes of hypercholesterolemia (e.g. poorly controlled diabetes mellitus, hypothyroidism, nephrotic syndrome, dysproteinasaemia, obstructive liver disease, other drug therapy, and alcoholism) should be identified and treated.

DOSEAGE AND ADMINISTRATION

For oral administration only.

The dosage range for Simvastatin is 5-80 mg/day, given as a single dose in the evening. Adjustments of dosage, if indicated, should be made at intervals of not less than 4 weeks, to a maximum of 80 mg/day given as a single dose in the evening. The 80mg dose is only recommended in patients at high risk for cardiovascular complications who have not achieved treatment goals on lower doses and when the benefits are expected to outweigh the potential risks.

The recommended usual starting dose is 20-40mg once a day in the evening.

Patients at High Risk of Coronary Heart Disease (CHD) or with Existing CHD

The usual starting dose of Simvastatin is 40mg/day given as a single dose in the evening in patients at high risk of CHD (with or without hyperlipidaemia), i.e., patients with diabetes, history of stroke or other cerebrovascular disease, peripheral vessel disease, or with existing CHD. Drug therapy can be initiated simultaneously with a standard cholesterol-lowering diet and exercise.

Patients with Hyperlipidaemia (Who Are Not in The Risk Categories Above)

The patient should be placed on a standard cholesterol-lowering diet before receiving Simvastatin and should continue on this diet during treatment with Simvastatin.

The usual starting dose is 20mg/day given as a single dose a day in the evening. Patients who require a large reduction in LDL-C (more than 45%) may be started at 40mg/day given as single dose in the evening. Patients with mild to moderate hypercholesterolemia can be treated with a starting dose of 10mg of Simvastatin. Adjustments of dosage, if required, should be made as specified above.

Homozygous Familial Hypercholesterolemia

The recommended dosage for patients with homozygous familial hypercholesterolemia is Simvastatin 40mg/day in the evening. The 80-mg dose is only recommended when the benefits are expected to outweigh the potential risks. Simvastatin should be used as an adjunct to other lipid-lowering treatments (e.g., LDL apheresis) in these patients if such treatments are unavailable.

In patients taking lomitapide concomitantly with Simvastatin, the dose of Simvastatin should not exceed 40 mg/day.

Concomitant Therapy

Simvastatin is effective alone or in combination with bile acid sequestrants.

In patient taking Simvastatin concomitantly with fibrates other than gemfibrozil or fenofibrate, the doses of Simvastatin should not exceed 10 mg/day. In patients taking atorvastatin, verapamil, diltiazem, or products containing elbasvir or grazoprevir concomitantly with Simvastatin the dose of Simvastatin should not exceed 20mg/day. In patients taking amiodipine or lipid-lowering dose of niacin (≥ 1 g/day) concomitantly with Simvastatin, the dose of Simvastatin should not exceed 40 mg/day.

Dosage in Renal Insufficiency

Because Simvastatin does not undergo significant renal excretion, modification of dosage should not be necessary in patients with moderate renal insufficiency.

In patients with severe renal insufficiency (creatinine clearance <30 mL/min), dosages above 10 mg/day should be carefully considered and, if deemed necessary, implemented cautiously.

Dosage in Pediatric Patients (10-17 Years of Age) with Heterozygous Familial Hypercholesterolemia

The recommended usual starting dose is 10 mg once a day in the evening. The recommended dosing range is 10-40mg/day; the maximum recommended dose is 40mg/day. Doses should be individualized according to the recommended goal of therapy.

CONTRAINDICATIONS

- Active liver disease or unexplained persistent elevations of serum transaminases.

- Pregnancy and lactation.

- Concomitant administration of potent CYP3A4 inhibitors (agents that increase AUC approximately 5 fold or greater) (e.g. itraconazole, ketconazole, posaconazole, voriconazole, erythromycin, clarithromycin, telithromycin, HIV protease inhibitors (e.g., neftravir), boceprevir, telaprevir, nefazodone and medicinal products containing cobicistat).

- Concomitant administration of gemfibrozil, ciclosporin, or danazol.

- In patients with homozygous familial hypercholesterolemia (HoFH), concomitant administration of lomitapide with doses > 40 mg simvastatin.

- Hypersensitivity to the active substance or to any of the excipients of the product.

WARNINGS AND PRECAUTIONS

There have been very rare reports of an immune-mediated necrotizing myopathy (IMNM) during or after treatment with some statins. IMNM is clinically characterized by:

- Persistent proximal muscle weakness and elevated serum creatine kinase, which persist despite discontinuation of statin treatment;
- Muscle biopsy showing necrotizing myopathy without significant inflammation;
- Improvement with immunosuppressive agents



Myopathy/Rhabdomyolysis

Simvastatin, like other inhibitors of HMG-CoA reductase, occasionally causes myopathy manifested as muscle pain, tenderness or weakness with creatine kinase (CK) above ten times the upper limit of normal (ULN). Myopathy sometimes takes the form of rhabdomyolysis with or without acute renal failure secondary to myoglobinuria, and very rare fatalities have occurred. The risk of myopathy is increased by high levels of HMG-CoA reductase inhibitory activity in plasma (i.e. elevated simvastatin and simvastatin acid plasma levels), which may be due, in part, to interacting drugs that interfere with simvastatin metabolism and/or transporter pathways.

As with other HMG-CoA reductase inhibitors, the risk of myopathy/rhabdomyolysis is dose related.

The risk of myopathy is greater in patients on simvastatin 80mg compared with other statin-based therapies with similar LDL-C-lowering efficacy.

Caution should be used when prescribing simvastatin to Asian patients and the lowest dose necessary should be employed.

-Reduced function of transport proteins

Reduced function of hepatic OATP transport proteins can increase the systemic exposure of simvastatin acid and increase the risk of myopathy and rhabdomyolysis. Reduced function can occur as the result of inhibition by interacting medicines (e.g. ciclosporin) or in patients who are carriers of the SLCO1B1 c.521T > C genotype.

Patients carrying the SLCO1B1 gene allele c.521T > C coding for a less active OATP1B1 protein have an increased systemic exposure of simvastatin acid and increased risk of myopathy.

-Creatine Kinase measurement

Creatine Kinase (CK) should not be measured following strenuous exercise or in the presence of any plausible alternative cause of CK increase as this makes value interpretation difficult. If CK levels are significantly elevated at baseline ($> 5 \times$ ULN), levels should be re-measured within 5 to 7 days later to confirm the results.

-Before the treatment

All patients starting therapy with simvastatin, or whose dose of simvastatin is being increased, should be advised of the risk of myopathy and told to report promptly any unexplained muscle pain, tenderness or weakness.

Caution should be exercised in patients with pre-disposing factors for rhabdomyolysis. In order to establish a reference baseline value, a CK level should be measured before starting a treatment in the following situations:

- Elderly (age ≥ 65 years).
- Female gender.
- Renal impairment.
- Uncontrolled hypothyroidism.
- Personal or familial history of hereditary muscular disorders.
- Previous history of muscular toxicity with a statin or fibrate.
- Alcohol abuse.

In such situations, the risk of treatment should be considered in relation to possible benefit, and clinical monitoring is recommended. If a patient has previously experienced a muscle disorder on a fibrate or a statin, treatment with a different member of the class should not be initiated with caution. If CK levels are significantly elevated at baseline ($> 5 \times$ ULN), treatment should not be started.

-Whilst on treatment

If muscle pain, weakness or cramps occur whilst a patient is receiving treatment with a statin, their CK levels should be measured. If these levels are found, in the absence of strenuous exercise, to be significantly elevated ($> 5 \times$ ULN), treatment should be stopped. If muscular symptoms are severe and cause daily discomfort, even if CK levels are $< 5 \times$ ULN, treatment discontinuation may be considered. If myopathy is suspected for any other reason, treatment should be discontinued.

A higher rate of myopathy has been observed in patients titrated to the 80mg dose. Periodic CK measurements are recommended as they may be useful to identify subclinical cases of myopathy. However, there is no assurance that such monitoring will prevent myopathy. Therapy with simvastatin should be temporarily stopped a few days prior to elective major surgery and when any major medical or surgical condition supervenes.

Headside effects

It is recommended that liver function tests be performed before treatment begins and thereafter when clinically indicated. Patients titrated to the 80mg dose should receive an additional test prior to titration, 3 months after titration to the 80mg dose, and periodically thereafter (e.g., semi-annually) for the first year of treatment. Special attention should be paid to patients who develop elevated serum transaminase levels, and in these patients, measurements should be repeated promptly and then performed more frequently. If the transaminase levels show evidence of progression, particularly if they rise to $3 \times$ ULN and are persistent, simvastatin should be discontinued. Note that ALT may emanate from muscle, therefore ALT rising with CK may indicate myopathy (see above Myopathy/Rhabdomyolysis).

There have been rare post-marketing reports of fatal and non-fatal hepatic failure in patients taking statins, including simvastatin. If serious liver injury with clinical symptoms and/or hyperbilirubinaemia or jaundice occur during treatment with simvastatin, promptly interrupt therapy. If an alternate etiology is not found, do not restart simvastatin.

The product should be used with caution in patients who consume substantial quantities of alcohol. As with other lipid-lowering agents, moderate ($< 3 \times$ ULN) elevations of serum transaminases have been reported following therapy with simvastatin. These changes appeared soon after initiation of therapy with simvastatin, were often transient, were not accompanied by any symptoms and interruption of treatment was not required.

Diabetes mellitus

Some evidence suggests that statins as a class raise blood glucose and in some patients, at high risk of future diabetes, may produce a level of hyperglycaemia where formal diabetes care is appropriate. This risk, however, is outweighed by the reduction in vascular risk with statins and therefore should not be a reason for stopping statin treatment. Patients at risk (fasting glucose 5.6 to 6.9 mmol/L, BMI > 30 kg/m², raised triglycerides, hypertension) should be monitored both clinically and biochemically according to national guidelines.

Interstitial lung disease

Cases of interstitial lung disease have been reported with some statins, including simvastatin, especially with long term therapy. Presenting features can include dyspnoea, non-productive cough and deterioration in general health (fatigue, weight loss and fever). If it is suspected a patient has developed interstitial lung disease, statin therapy should be discontinued.

Myasthenia Gravis/Ocular Myasthenia

In few cases, statins have been reported to induce de novo or aggravate pre-existing myasthenia gravis or ocular myasthenia. Covastin Tablet should be discontinued in case of aggravation of symptoms. Recurrences when the same or a different statin was re-administered have been reported.

Excipient

This product contains lactose. Patients with rare hereditary problems of galactose intolerance, the Lapp lactase deficiency or glucose-galactose malabsorption should not take this medicine.

DRUG INTERACTIONS

Drug Interactions Associated with Increased Risk of Myopathy/Rhabdomyolysis		
Interacting agents	Prescribing recommendations	
Potent CYP3A4 inhibitors, e.g.: Itraconazole, Ketconazole, Posaconazole, Voriconazole, Erythromycin, Clarithromycin, Telithromycin, HIV protease inhibitors (e.g., neftravir), Boceprevir, Telaprevir, Nefazodone, Cobicistat, Ciclosporin, Danazol, Gemfibrozil	Contraindicated with simvastatin	
Other fibrates (except fenofibrate)	Do not exceed 10mg simvastatin daily	
Fusidic acid	Is not recommended with simvastatin	
Niacin (nicotinic acid) (≥ 1 g/day)	For Asian patients, not recommended with simvastatin	
Amiodarone, Amlodipine, Verapamil, Diltiazem, Elbasvir, Grazoprevir	Do not exceed 20mg simvastatin daily	
Lomitapide	For patients with HoFH, do not exceed 40mg simvastatin daily	
Daptomycin	It should be considered to temporarily suspend simvastatin in patients taking daptomycin unless the benefits of concomitant administration outweigh the risk	
Grapefruit juice	Avoid grapefruit juice when taking simvastatin	

Contraindicated Drugs

Potent inhibitors of CYP3A4: Concomitant use with medicines labelled as having a potent inhibitory effect on CYP3A4 at therapeutic doses (e.g.: itraconazole, ketconazole, posaconazole, voriconazole, erythromycin, clarithromycin, telithromycin, HIV protease inhibitors (e.g. neftravir), boceprevir, telaprevir or nefazodone) is contraindicated. If treatment with potent CYP3A4 inhibitors is unavoidable, therapy with simvastatin should be suspended during the course of treatment.

Gemfibrozil, cyclosporine or danazol: concomitant use of these drugs with simvastatin is contraindicated.

Interactions involving inhibitors of CYP3A4

Simvastatin is a substrate of cytochrome P450 3A4. Potent inhibitors of cytochrome P450 3A4 increase the risk of myopathy and rhabdomyolysis by increasing the concentration of HMG-CoA reductase inhibitory activity in plasma during simvastatin therapy. Such inhibitors include itraconazole, ketconazole, posaconazole, voriconazole, erythromycin, clarithromycin, telithromycin, HIV protease inhibitors (e.g. neftravir), boceprevir, telaprevir, nefazodone and medicinal products containing cobicistat. Concomitant administration of itraconazole resulted in a more than 10-fold increase in exposure to simvastatin acid (the active beta-hydroxyacid metabolite). Telithromycin caused an 11-fold increase in exposure to simvastatin acid.

Combination with itraconazole, ketconazole, posaconazole, voriconazole, HIV protease inhibitors (e.g. neftravir), boceprevir, telaprevir, erythromycin, clarithromycin, telithromycin, nefazodone and medicinal products containing cobicistat are contraindicated, as well as gemfibrozil, ciclosporin, and danazol. If treatment with potent CYP3A4 inhibitors (agents that increase AUC approximately 5 fold or greater) is unavoidable, therapy with simvastatin must be suspended (and use of an alternative statin considered) during the course of treatment. Caution should be exercised when combining simvastatin with certain other less potent CYP3A4 inhibitors: fluconazole, verapamil, or diltiazem. If treatment with potent CYP3A4 inhibitors is unavoidable, therapy with simvastatin must be suspended (and use of an alternative statin considered) during the course of treatment.

Moderate inhibitors of CYP3A4

Patients taking other medicines labelled as having a moderate inhibitory effect on CYP3A4 concomitantly with simvastatin, particularly higher simvastatin doses, may have an increased risk of myopathy.

When co-administering simvastatin with a moderate inhibitor of CYP3A4 (agents that increase AUC approximately 2-5 fold), a dose adjustment of simvastatin may be necessary. For certain moderate CYP3A4 inhibitors e.g. diltiazem, a maximum dose of 20mg simvastatin is recommended.

Fluconazole

Rare cases of rhabdomyolysis associated with concomitant administration of simvastatin and fluconazole have been reported.

Calcium Channel Blockers

-Verapamil or Diltiazem

In a clinical trial, patients on diltiazem treated concomitantly with simvastatin 80 mg had an increased risk of myopathy. The dose of simvastatin should not exceed 20 mg daily in patients receiving concomitant medication with verapamil or diltiazem.

-Amlodipine

In a clinical trial, patients on amlodipine treated concomitantly with simvastatin 80 mg had a slightly increased risk of myopathy. The dose of simvastatin should not exceed 40 mg daily in patients receiving concomitant medication with amlodipine.

-Niacin (≥ 1 g/day)

The dose of simvastatin should not exceed 40mg daily in patients receiving concomitant medication with niacin (nicotinic acid) ≥ 1 g/day. Cases of myopathy/rhabdomyolysis have been observed with simvastatin co-administered with lipid-modifying doses (≥ 1 g/day) of niacin.

Ciclosporin

The risk of myopathy/rhabdomyolysis is increased by concomitant administration of ciclosporin with simvastatin; therefore, use with ciclosporin is contraindicated. Although the mechanism is not fully understood, ciclosporin has been shown to increase the AUC of HMG-CoA reductase inhibitors. The increase in AUC for simvastatin acid is presumably due, in part, to inhibition of CYP3A4 and/or OATP1B1.

Danazol

The risk of myopathy and rhabdomyolysis is increased by concomitant administration of danazol with simvastatin; therefore, use with danazol is contraindicated.

Gemfibrozil

Gemfibrozil increases the AUC of simvastatin by 1.9-fold, possibly due to inhibition of the glucuronidation pathway and/or OATP1B1. Concomitant administration with gemfibrozil is contraindicated.

Fusidic acid

Simvastatin must not be co-administered with systemic formulations of fusidic acid or within 7 days of stopping fusidic acid treatment. In patients where the use of systemic fusidic acid is considered essential, statin treatment should be discontinued throughout the duration of fusidic acid treatment. There have been reports of rhabdomyolysis (including some fatalities) in patients receiving fusidic acid and statins in combination. The patient should be advised to seek medical advice immediately if they experience any symptoms of muscle weakness, pain or tenderness. Statin therapy may be re-initiated seven days after the last dose of fusidic acid. In exceptional circumstances, where prolonged systemic fusidic acid is needed, e.g. for the treatment of severe infections, the need for co-administration of simvastatin and fusidic acid should only be considered on a case by case basis and under close medical supervision.

Amiodarone

The risk of myopathy and rhabdomyolysis is increased by concomitant administration of amiodarone with simvastatin. In a clinical trial, Myopathy was reported in 6% of patients receiving simvastatin 80mg and amiodarone. Therefore, the dose of simvastatin should not exceed 20mg daily in patients receiving concomitant medication with amiodarone.

Lomitapide

The risk of myopathy and rhabdomyolysis may be increased by concomitant administration of lomitapide with simvastatin. Therefore, in patients with HoFH, the dose of simvastatin must not exceed 40mg daily in patients receiving concomitant medication with lomitapide.

Inhibitors of the Transport Protein OATP1B1

Simvastatin acid is a substrate of the transport protein OATP1B1. Concomitant administration of medicinal products that are inhibitors of the transport protein OATP1B1 may lead to increased plasma concentrations of simvastatin acid and an increased risk of myopathy.

Inhibitors of Breast Cancer Resistant Protein (BCRP)

Simvastatin is a substrate of the Breast Cancer Resistant Protein (BCRP) efflux transporter. Concomitant administration of products that are inhibitors of BCRP (e.g., elbasvir and grazoprevir) may lead to increased plasma concentrations of simvastatin and an increased risk of myopathy; therefore, a dose adjustment of simvastatin should be considered depending on the prescribed dose. Co-administration of elbasvir and grazoprevir with simvastatin has not been studied; however, the dose of simvastatin should not exceed 20mg daily in patients receiving concomitant medication with products containing elbasvir or grazoprevir.

Acipimox

Acipimox is structurally related to niacin. Although acipimox was not studied, the risk for muscle related toxic effects may be similar to niacin.

Grapefruit juice

Grapefruit juice inhibits cytochrome P450 3A4. Concomitant intake of large quantities (over 1 litre daily) of grapefruit juice and simvastatin resulted in a 7-fold increase in exposure to simvastatin acid. Intake of 240 mL of grapefruit juice in the morning and simvastatin in the evening also resulted in a 1.9-fold increase. Intake of grapefruit juice during treatment with simvastatin should therefore be avoided.

Colchicine

There have been reports of myopathy and rhabdomyolysis with the concomitant administration of colchicine and simvastatin in patients with renal impairment. Close clinical monitoring of such patients taking this combination is advised.

Daptomycin

Cases of myopathy and/or rhabdomyolysis have been reported with HMG-CoA reductase inhibitors (e.g. simvastatin) coadministered with daptomycin. Caution should be used when prescribing HMG-CoA reductase inhibitors with daptomycin, as either agent can cause myopathy and/or rhabdomyolysis when given alone. Consideration should be given to temporarily suspend simvastatin in patients taking daptomycin unless the benefits of concomitant administration outweigh the risk. Consult the prescribing information of daptomycin to obtain further information about this potential interaction with HMG-CoA reductase inhibitors (e.g. simvastatin) and for further guidance related to monitoring.

Rifampicin

Because rifampicin is a potent CYP3A4 inducer, patients undertaking long-term rifampicin therapy (e.g. treatment of tuberculosis) may experience loss of efficacy of simvastatin. In a pharmacokinetic study in normal volunteers, the area under the plasma concentration curve (AUC) for simvastatin acid was decreased by 93% with concomitant administration of rifampicin.

Effects of simvastatin on the pharmacokinetics of other medicinal products

Simvastatin does not have an inhibitory effect on cytochrome P450 3A4. Therefore, simvastatin is not expected to affect plasma concentrations of substances metabolised via cytochrome P450 3A4.

Oral anticoagulants

Very rare cases of elevated INR have been reported. In patients taking coumarin anticoagulants, prothrombin time should be determined before starting simvastatin and frequently enough during early therapy to ensure that no significant alteration of prothrombin time occurs. Once a stable prothrombin time has been documented, prothrombin times can be monitored at the intervals usually recommended for patients on coumarin anticoagulants. If the dose of simvastatin is changed or discontinued, the same procedure should be repeated. Simvastatin therapy has not been associated with bleeding or with changes in prothrombin time in patients not taking anticoagulants.

Other Drugs

Other fibrates: The dose of simvastatin should not exceed 10 mg daily in patients receiving concomitant medication with fibrates other than gemfibrozil or fenofibrate. When simvastatin and fenofibrate are given concomitantly, there is no evidence that the risk of myopathy exceeds the sum of individual risks of each agent. Caution should be used when prescribing fenofibrate with simvastatin, as either agent can cause myopathy when given alone. Addition of fibrates to simvastatin typically provides little additional reduction in LDL-C, but further reductions of TG and further increases in HDL-C may be obtained. Combinations of fibrates with simvastatin have been used without myopathy in small short-term clinical studies with careful monitoring.

PREGNANCY AND LACTATION

Pregnancy

Simvastatin is contraindicated during pregnancy. Safety in pregnant women has not been established. No controlled clinical trials with simvastatin have been conducted in pregnant women. Rare reports of congenital anomalies following intrauterine exposure to HMG-CoA reductase inhibitors have been received.

Atherosclerosis is a chronic process, and ordinarily discontinuation of lipid-lowering medicinal products during pregnancy should have little impact on the long-term risk associated with primary hypercholesterolemia. For these reasons, simvastatin must not be used in women who are pregnant, trying to become pregnant or suspect they are pregnant. Treatment with simvastatin must be suspended for the duration of pregnancy or until it has been determined that the woman is not pregnant.

Lactation

It is not known whether simvastatin or its metabolites are excreted in human milk. Because many medicinal products are excreted in human milk and because of the potential for serious adverse reactions, women taking simvastatin must not breastfeed their infants.

SIDE EFFECTS

Blood and lymphatic system disorders:

Rare: anaemia

Immune system disorders:

Very rare: anaphylaxis

Psychiatric disorders:

Very rare: insomnia

Not known: depression